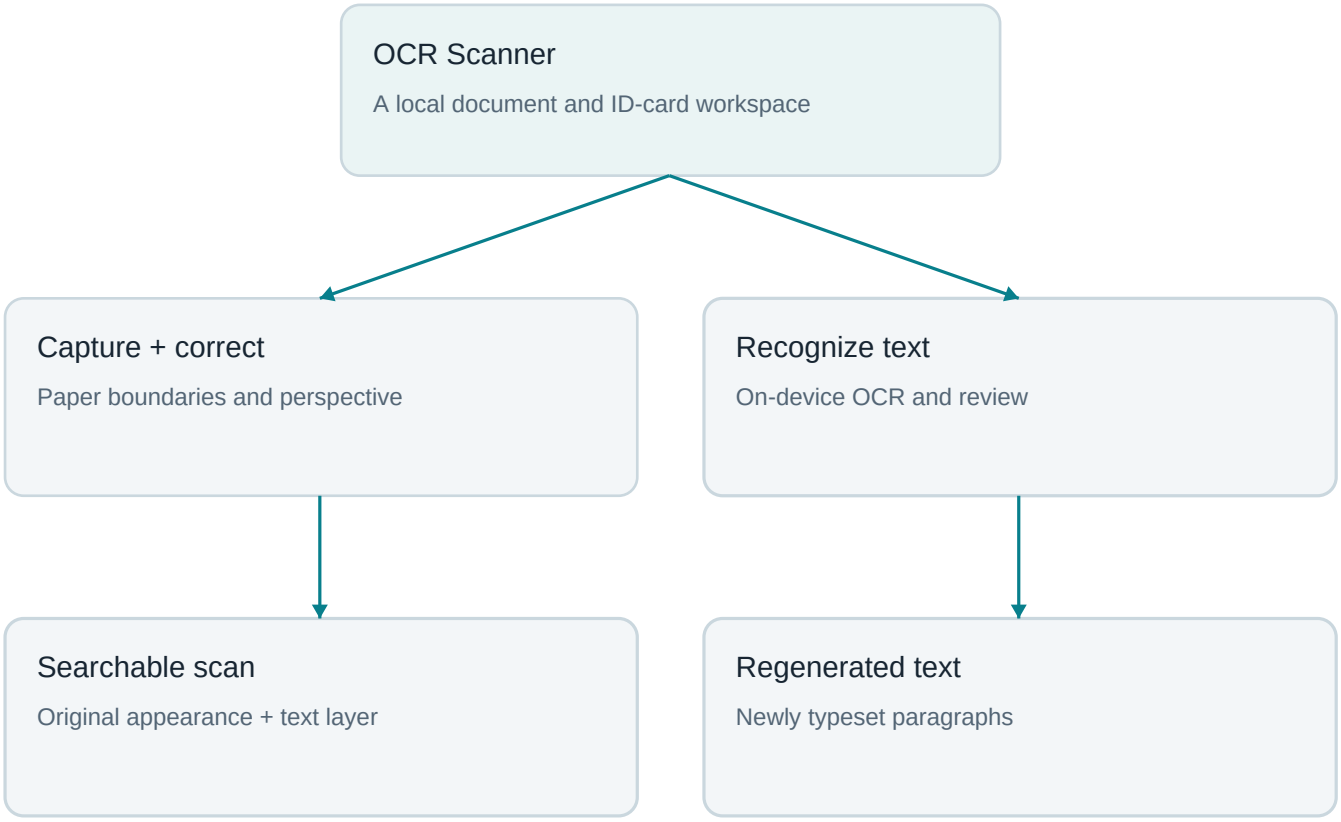
What we are designing

15 screens: 14 product screens + 1 optional computer vision learning lab.



Confirmed scope

Both PDF formats. Front/back card pairing. Multiple cards in one photograph. Arrange cards on the same PDF page.

Design baseline

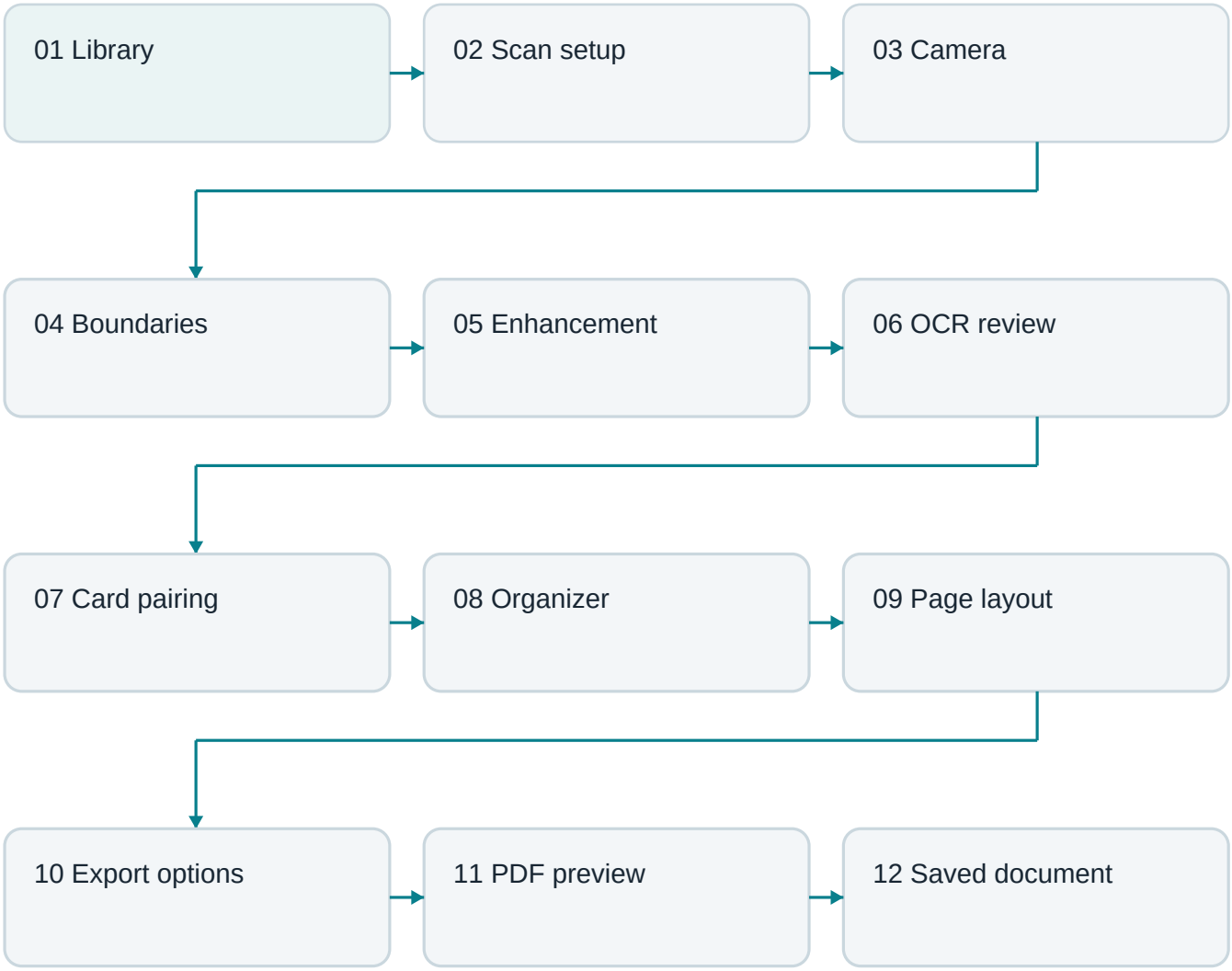
Kotlin + Compose. CameraX + OpenCV. Bundled ML Kit OCR. Room and private files. Original captures and editable drafts retained.

Later scope

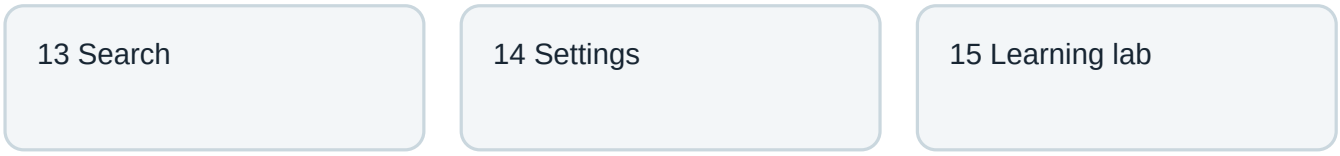
Cloud sync, handwriting, curved-page flattening, automatic card pairing, and exact reconstruction of complex documents.

Screen connections

Main routes below. System dialogs, sheets, and processing states do not add screens.



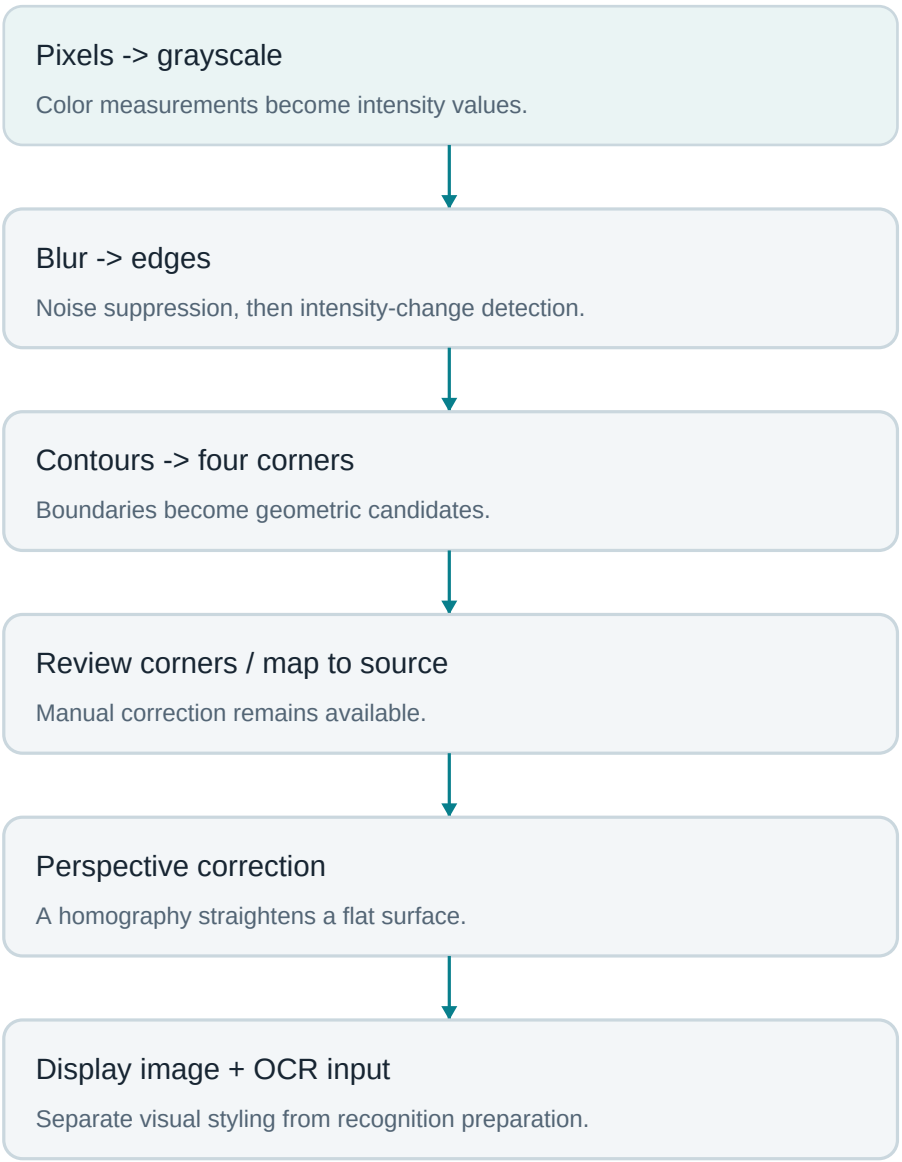
Document mode skips 07. Import goes from 02 directly to 04. Saved items open 12; Edit returns to 08. Opening a saved PDF reuses 11.



01 opens Search and Settings. 05 opens Learning lab and returns with edits preserved.

How scanning works

Expose each stage in the learning lab. Use full-resolution captures for the final correction.



Camera strategy

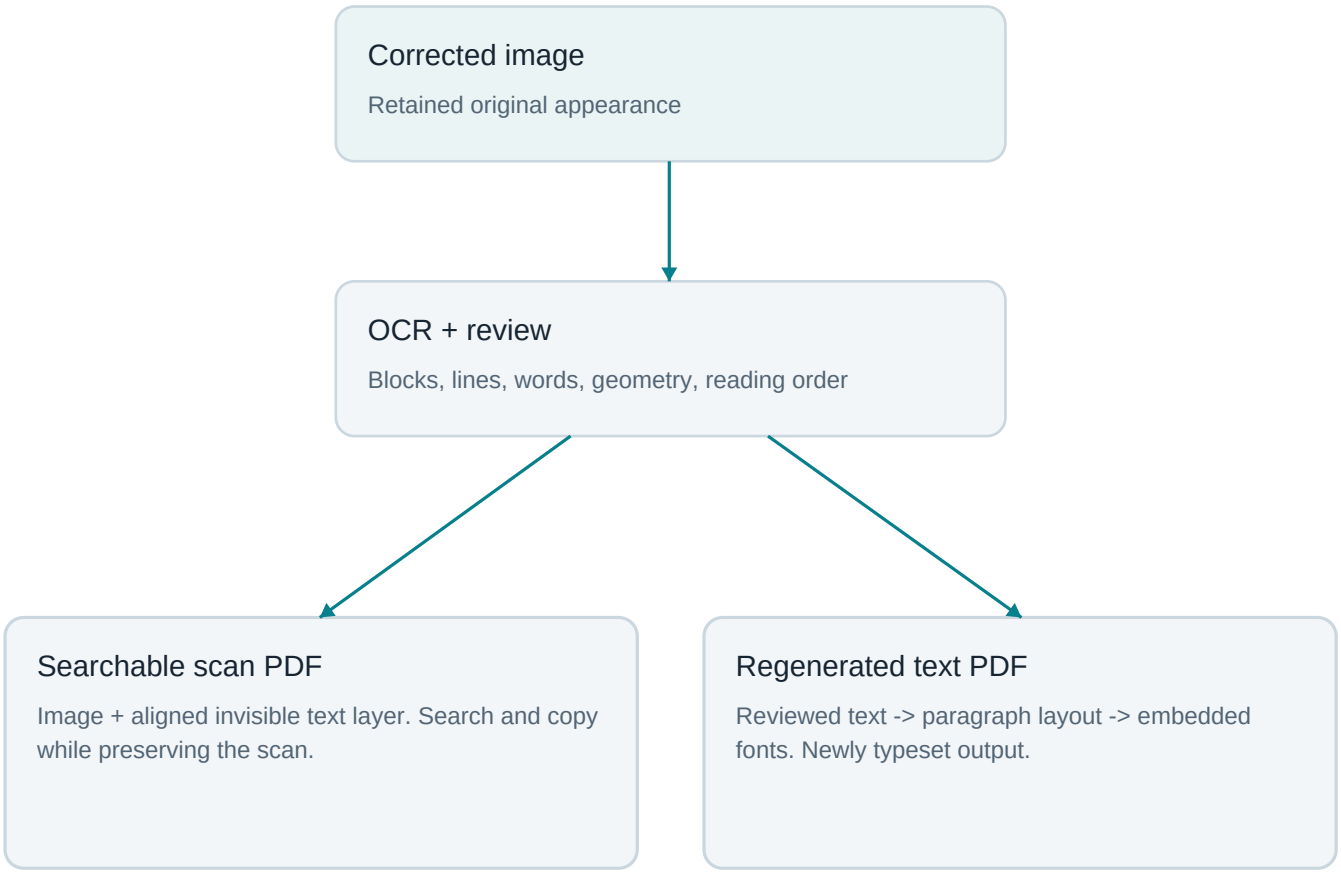
Analyze reduced preview frames for guidance. Capture at higher resolution. Close analysis frames promptly; keep full OCR out of the live preview loop.

Failure behavior

Glare, blur, clutter, and shadows can hide boundaries or text. Offer recapture and manual corners; filters cannot recreate missing detail.

OCR and PDF outputs

OCR produces words and positions. Export decides how they appear on the page.



Alignment contract

Use a shared transform for image and text: rotation -> crop -> corrected region -> PDF placement. Keep text revisions tied to image revisions.

Editing contract

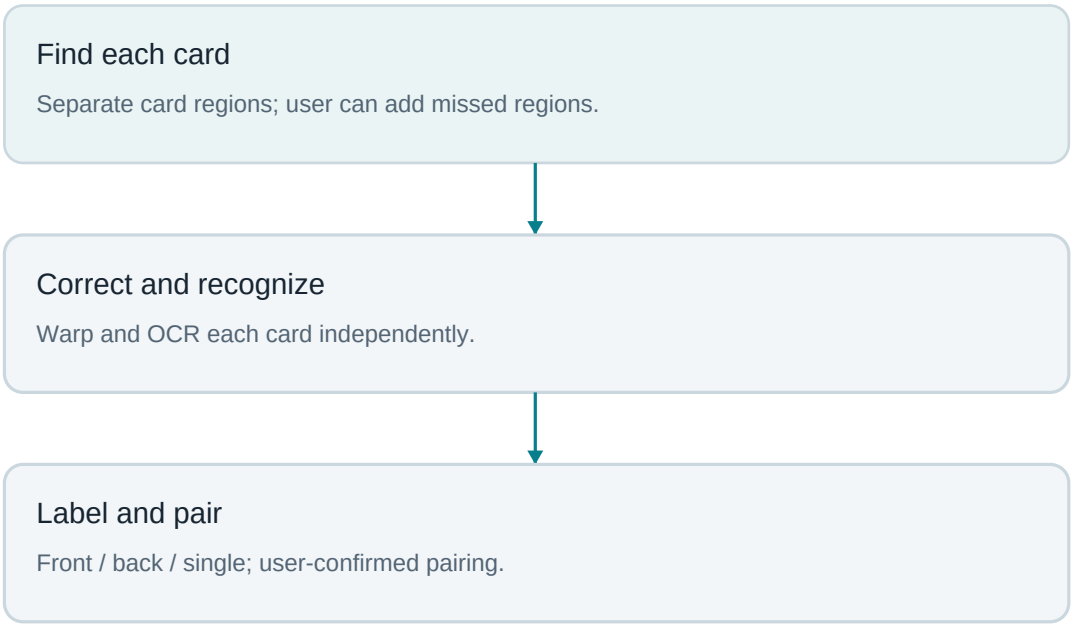
Short word edits preserve geometry where possible. Large rewrites require alignment review for searchable scans. Regenerated text can reflow.

PDF-engine gate

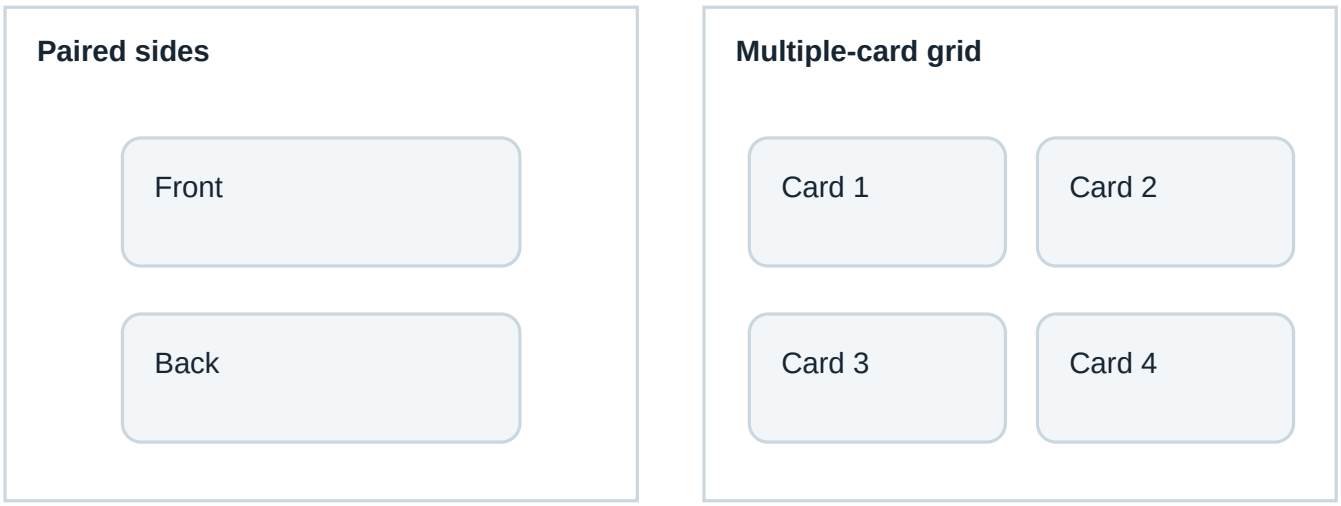
Verify invisible text, Unicode mapping, font embedding, extraction order, and Android compatibility in two readers before choosing a library. Android PdfDocument is a candidate, not a proven searchable exporter.

ID cards: capture to page

Detect regions, read text, confirm labels and pairs, then compose the export sheet.



PAGE TEMPLATES

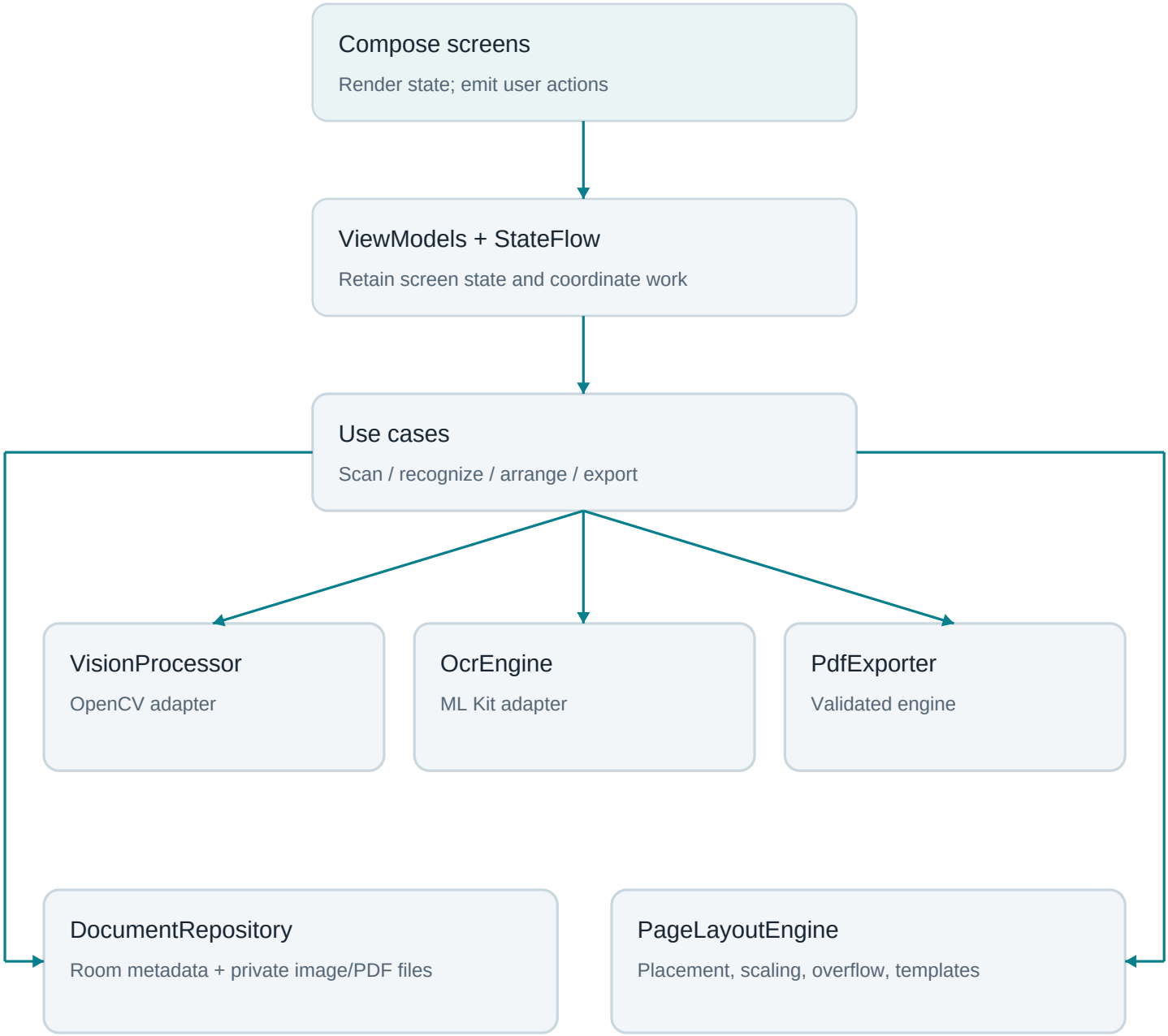


Preserve aspect ratio. Use physical size only when known. Overflow creates another page. Card imagery belongs to scan layouts; regenerated output uses a separate text layout.

Initial capture target: fully visible, separated cards. Unknown document types remain valid; no automatic identity or authenticity verification.

Kotlin application architecture

One app module initially. Keep processing, presentation, and persistence behind clear interfaces.



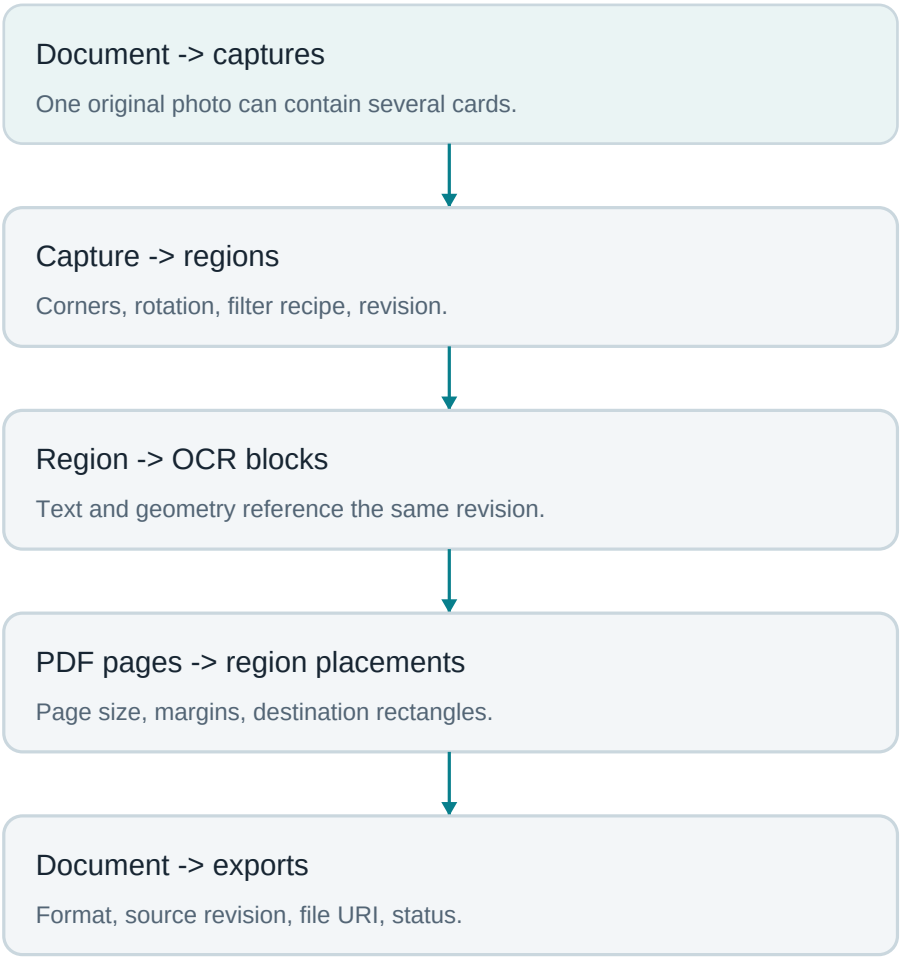
Execution

CameraX captures images. Coroutines keep interactive work off the UI thread. WorkManager handles durable OCR/export batches, not preview frames.

Packages: feature/{library,capture,editor,ocr,cards,export}, core/{vision,ocr,pdf}, data.

Data and revision lifecycle

Image files are separate from metadata. Each result belongs to a specific document revision.



Edit invalidation

Crop or rotation change -> rerun OCR and regenerate PDF. Layout-only change -> regenerate PDF while retaining OCR. Never let an old job overwrite a newer edit.

Draft and export lifecycle

Captured -> corrected -> recognized -> reviewed -> arranged -> exported. Failures preserve the last valid draft and saved export; users can retry or cancel.

Storage

Room: documents, recipes, OCR, pairs, placements. Private files: originals, derivatives, PDFs. Deleting a document removes app-owned files; external shared copies are separate.

Screen specifications / 1 of 3

Stable route IDs connect the design frames, prototype, and future Kotlin navigation.

S01 Document library

Find saved documents and resume drafts.

Show: Thumbnails; title; page count; draft/export status; New scan; Search; Settings.

Connect: New scan -> 02; item -> 12; Search -> 13; Settings -> 14.

Variants: Empty library; populated library; draft with interrupted work; selection/delete confirmation.

S02 Scan setup

Choose the scan workflow and input source.

Show: Document or ID cards; ID workflow: front/back or multiple cards; Camera or Import images.

Connect: Camera -> 03; system image picker -> 04; Back -> caller.

Variants: Document; front/back cards; multiple cards; import cancelled. Preserve the caller and current draft.

S03 Camera capture

Capture sharp pages or cards with boundary guidance.

Show: Viewfinder; shutter; flash; capture count; thumbnail tray; Done; optional front/back cue.

Connect: Done -> 04; retake returns to its region; Back -> draft/discard sheet.

Variants: No permission; no boundary; blur/glare suggestion; captured images; missing back. Suggestions must not block manual capture.

S04 Boundary editor

Confirm the page or each card region before correction.

Show: Four draggable corners; zoom; rotate; add/remove region in multiple-card mode; image switcher; Apply.

Connect: Apply -> 05; Retake -> 03; Cancel -> caller.

Variants: Detected corners; no detection/manual corners; multiple cards; invalid/crossed corners. Keep edits per capture.

S05 Enhancement editor

Choose a readable appearance while preserving the original.

Show: Original/color/grayscale/B&W thumbnails; brightness/contrast; rotate; before/after; Apply; Learning lab.

Connect: Continue -> OCR processing -> 06; Lab -> 15; Back -> 04 or caller.

Variants: Original; color; B&W; clipped highlights; processing failure. Apply per region; batch apply is explicit.

Screen specifications / 2 of 3

Stable route IDs connect the design frames, prototype, and future Kotlin navigation.

S06 OCR review

Review extracted text against the scan.

Show: Image/Text tabs; word highlight; block editor; language; rerun; previous/next region; Continue.

Connect: Document Continue -> 08; ID Continue -> 07; selected word -> edit sheet.

Variants: Recognizing; text found; no text; unsupported script; edits; retry. Do not invent confidence percentages.

S07 Card review and pairing

Label card regions and confirm front/back relationships.

Show: Card thumbnails; generic/unknown type label; Front/Back/Single; pair/unpair; capture missing side; Continue.

Connect: Continue -> 08; capture missing side -> 02 with pending pair; edit text -> 06.

Variants: Unpaired; paired; single-sided; missing back; unknown type. Manual confirmation is authoritative.

S08 Document organizer

Manage the complete draft before export.

Show: Title; page/card thumbnails; drag reorder; add; crop; enhance; text; duplicate; delete; pair cards in ID mode; Layout.

Connect: Editors -> 04/05/06; pairing -> 07; Add -> 02; Layout -> 09.

Variants: One page; multipage; multiple card pairs; empty after deletion; stale OCR after crop change.

S09 Page layout

Place pages and cards on export sheets.

Show: Page canvas; A4/Letter/Auto; margins; orientation; fit; ID templates: stacked, side-by-side, grid; overflow page count.

Connect: Continue -> 10; Back -> 08.

Variants: Document fit; card pair; card grid; overflow; unknown physical size. Preserve aspect ratio; use actual size only when known.

S10 Export options

Choose what the generated PDF contains.

Show: Searchable scan or Regenerated text; filename; output summary; scan compression or text font/size; Generate preview.

Connect: Generate -> progress state -> 11; Adjust layout -> 09; incomplete OCR -> 06.

Variants: Both output modes; pending OCR; failed OCR. Text mode says simple paragraphs; no exact-layout promise.

Screen specifications / 3 of 3

Stable route IDs connect the design frames, prototype, and future Kotlin navigation.

S11 PDF preview

Inspect the generated result before saving or sharing.

Show: Page viewer; page counter; search; text selection; output-mode label; Save; change options; edit layout.

Connect: Save -> 12; Change options -> 10; layout -> 09; Back -> caller.

Variants: Generating overlay; preview ready; no search match; failed export; outdated preview. Shared reader UI for both formats.

S12 Saved document details

Open, edit, export, or share a saved document.

Show: Title; cover; page count; exports with format/time; Open PDF; Edit; Review text; Share; rename/delete menu.

Connect: Open -> 11 read-only saved mode; Edit -> 08; Text -> 06; Share -> system share sheet; Back -> 01.

Variants: Draft only; current export; older export clearly labeled; export missing; failed share. Regenerate only on request.

S13 Search results

Find documents using titles and recognized text.

Show: Query field; text matches; highlighted snippets; document/page labels; recent query optional.

Connect: Result -> 12 with matched page context; Back -> 01.

Variants: Typing; results; no matches; OCR not run. Searches local metadata and OCR only.

S14 Settings

Set scanning and export defaults.

Show: Theme; OCR script choice; default paper size; scan quality; learning-lab visibility; local-storage summary.

Connect: Save defaults / Back -> 01.

Variants: Default values; changed defaults; unsupported language. Only expose supported OCR scripts; no model-training UI.

S15 CV learning lab

Inspect how image processing creates the scan.

Show: Stage selector: original, grayscale, blur, edges, contours, corners, warp, OCR boxes; two-image comparison; short explanation.

Connect: Back -> 05 with edits preserved.

Variants: Stage result; detection failure; no OCR yet; comparison. Experiments never alter the document until explicitly applied.

Design handoff and gates

Finalize visual design before implementing the application.

Design deliverables

15 primary frames + named state variants. Design tokens and reusable components. Connected prototype covering both PDF formats and both card workflows. Android Back and editor return paths annotated.

Shared states, not extra screens

Camera permission, image picker, share sheet, progress, word editor, rename, delete confirmation, draft/discard, and failure/retry sheets.

Acceptance checks

No dead-end primary actions. Drafts survive navigation. No clipped content or distorted cards. Empty/error states have recovery actions. Both previews show the correct output type. Test compact widths and enlarged text.

Build sequence after design

Pixels and filters -> document geometry -> OCR -> PDF compatibility prototype -> document workflow -> card layouts -> Android reliability.

Decisions requiring implementation validation

Exact PDF engine, target Android/device baseline, additional writing scripts, named card families, physical-size printing, and measured quality/performance targets. These do not block the generic frontend design.

OFFICIAL REFERENCES

Android architecture

CameraX

OpenCV Android

ML Kit OCR

Android PDF pages